Sachin Yadav

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Education

Cornell University *August 2026 – Present*
Ph.D. in Computer Science

Indian Institute of Technology (IIT) Gandhinagar *2018 – 2022*
B.Tech. in Computer Science and Engineering
GPA: 9.26/10.0

Work Experience

Google DeepMind *October 2024 – August 2026*
Pre-Doctoral Researcher in Frontier AI Unit
Advisors: Dr. Dheeraj Nagaraj, Dr. Karthikeyan Shanmugam

- Formulated and implemented a novel pre-training objective for diffusion language models, driving **substantial improvements in validation metrics**; **adopted by the Gemini Diffusion team**, resulting in a **provisional patent** and a **Spot Bonus**.
- Developed **Interleaved Gibbs Diffusion (IGD)**, a unified discrete-continuous diffusion framework for constrained generation (ICLR DeLTa Workshop 2025; provisional patent).
- Conducted research on large-scale training and inference efficiency for LLMs; contributor to **Gemini 2.5** technical report.

Microsoft Research India *July 2024 – October 2024*
Research Intern
Advisor: Dr. Amit Sharma, Principal Researcher
Worked on enhancing the causal reasoning capabilities of large language models by integrating causal analysis tools within an agent-based system.

Microsoft Research India *July 2022 – July 2024*
Research Fellow (AI Resident) in Machine Learning and AI Group
Advisor: Dr. Manik Varma, Technical Fellow & Corporate Vice President
Developed large-scale machine learning models for extreme multi-label classification (industry-scale search, retrieval and recommendation systems), leading to top-tier publications and impact across Microsoft products.

Publications

* - equal contribution

1. **Gemini 2.5 Technical Report**.
   Gemini Team (including **Sachin Yadav**), Google DeepMind
   ***arXiv** Preprint, 2025*

2. **Interleaved Gibbs Diffusion for Constrained Generation**.
   Gautham Govind Anil, **Sachin Yadav**, Dheeraj Nagaraj, Karthikeyan Shanmugam, Prateek Jain
   *International Conference on Learning Representations (ICLR) (**DeLTa Workshop**), 2025*

3. **On the Necessity of World Knowledge for Mitigating Missing Labels in Extreme Classification**.
   Jatin Prakash*, Anirudh Buvanesh*, Bishal Santra, Deepak Saini, **Sachin Yadav**, Jian Jiao, Yashoteja Prabhu, Amit Sharma, Manik Varma
   *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2025*

4. **Extreme Meta-Classification for Large-Scale Zero-Shot Retrieval**.
   Sachin Yadav^{+}*****, Deepak Saini*, Anirudh Buvanesh*, Bhawna Paliwal, Kunal Dahiya, Siddarth Asokan, Yashoteja Prabhu, Jian Jiao, Manik Varma
   $^{+}$ - led the project
   *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2024*

5. **Deep Encoders with Auxiliary Parameters for Extreme Classification**.
   Kunal Dahiya, **Sachin Yadav**, Sushant Sondhi, Deepak Saini, Sonu Mehta, Jian Jiao, Sumeet Agarwal, Purushottam Kar, Manik Varma
   *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2023*

6. **Deep Gaussian Processes for Air Quality Inference**. Extended Abstract
   Aadesh Desai*, Eshan Gujarathi*, Saagar Parikh*, **Sachin Yadav***, Zeel Patel, and Nipun Batra
   *Young Researchers' Symposium @ Joint International Conference on Data Science & Management of Data (**CODS-COMAD**), 2023*

Research Adoption

‡ - Specific details withheld due to non-disclosure agreements.

1. **Gemini Diffusion: State-of-the-art text diffusion model from Google DeepMind** ‡
   Formulated and implemented a novel pre-training objective for the text diffusion model, driving substantial improvements in validation metrics. Awarded a **Spot Bonus** for delivering technical advancements to the project.

2. **Search Engine: Sponsored Ads** ‡
   EMMETT: Extreme Meta-Classification for Large-Scale Zero-Shot Retrieval.
   On live traffic from a major search engine, EMMETT resulted in a 4.2% increase in Click-Through Rate (CTR) and a 0.9% reduction in Quick Back Rate (QBR) for Sponsored Ads.

3. **Personalized Ad Recommendations** ‡
   EMMETT: Extreme Meta-Classification for Large-Scale Zero-Shot Retrieval.
   In live trials on a major display ads platform, EMMETT achieved a 3.3% improvement in Click-Through Rate (CTR) and a 4.9% improvement in Click Yield for personalized ads.

Patents

1. **Discrete Diffusion Neural Networks with Planning**. 2025. (Provisional Patent Filed, Google DeepMind).
   Inventors: **Sachin Yadav**, Dheeraj Nagaraj, Karthikeyan Shanmugam

2. **Interleaved Gibbs Diffusion for Constrained Generation**. 2025. (Provisional Patent Filed, Google DeepMind).
   Inventors: Dheeraj Nagaraj, Karthikeyan Shanmugam, **Sachin Yadav**, Gautham Govind Anil, Prateek Jain

3. **Retrieval of Novel Keywords for Search**. 2025. (Granted, Microsoft, US Patent 12499168).
   Inventors: Deepak Saini, Jian Jiao, **Sachin Yadav**, Anirudh Buvanesh, Bhawna Paliwal, Manik Varma

Selected Awards and Honors

- Awarded a Spot Bonus at Google DeepMind for technical contributions to Gemini Diffusion. *2025*
- Awarded the Ashok Jain Scholarship for merit, given to one B.Tech. student yearly at IIT Gandhinagar. *2021 & 2022*
- Selected for the ACM-ICPC Regionals programming competition. *2019*
- Featured in Dean's List for semesters 1, 2, and 3 for exceptional academic performance at IIT Gandhinagar.
- Awarded the KVPY Fellowship, a national science fellowship by the Government of India. *2016*

Selected Projects

Gemini Text Diffusion
Advisors: Dr. Dheeraj Nagaraj, Dr. Karthikeyan Shanmugam, Dr. Prateek Jain, Dr. Brendan O'Donoghue, *Google DeepMind* — [Provisional Patent Filed, 2025]

- Contributed to Gemini Diffusion, a diffusion-based LLM that generates text via iterative noise refinement, matching state-of-the-art performance of large autoregressive models.
- Led innovations in the pre-training objective for Gemini Diffusion, resulting in significant quality gains in text generation.

Interleaved Gibbs Diffusion (IGD): Generating Discrete-Continuous Data with Implicit Constraints
Advisors: Dr. Dheeraj Nagaraj, Dr. Karthikeyan Shanmugam, Dr. Prateek Jain, *Google DeepMind* — [ICLR DeLTa Workshop 2025]

- Introduced Interleaved Gibbs Diffusion (IGD), a generalized Markov chain framework that theoretically guarantees exact reversal of forward processes and flexibly combines discrete and continuous denoisers for robust hybrid data generation.
- Empirically demonstrated significant out-of-the-box improvements in constrained tasks like 3-SAT, showcasing IGD's ability to handle implicit constraints in logical reasoning without domain-specific adaptations.

- Achieved leading performance on molecular structure, layout, and tabular data generation tasks using IGD, relying solely on general modeling strategies rather than task-specific inductive biases.

EMMETT: Extreme Meta-Classification for Large-Scale Zero-Shot Retrieval
Advisor: Dr. Manik Varma, *Microsoft Research India* — [KDD 2024]

- Developed EMMETT, an innovative framework for synthesizing classifiers for novel items in zero-shot retrieval, enhancing both accuracy and efficiency.
- Introduced IRENE, an effective instantiation of EMMETT suited for large-scale deployments, achieving up to 15% improvement in zero-shot retrieval accuracy and seamlessly integrating with existing Siamese encoders.
- Formulated a theoretical framework to guide algorithm and training strategy design for large-scale zero-shot retrieval, ensuring robust generalization across diverse applications.
- Empirically demonstrated the effectiveness of IRENE through comprehensive experiments and online A/B testing in a major search engine, resulting in a 4.2% increase in ad click-through rate.

DEXA: Deep Encoders with Auxiliary Parameters for Extreme Classification
Advisors: Prof. Purushottam Kar and Dr. Manik Varma, *Microsoft Research India* — [KDD 2023]

- Identified semantic gap and data paucity issues in Extreme Classification settings, leading to suboptimal encoder training.
- Proposed DEXA, a lightweight alternative that enhances Encoder embeddings with extra auxiliary parameters on the label side, addressing the identified training issues.
- Conducted comprehensive experiments with various architectures, showcasing DEXA's modularity within existing XC solutions, all achieved with minimal time and memory overheads during training.
- Attained accuracy improvements of 6% and 15% on public and proprietary benchmarks respectively.